SAYED MASHEQUL BARI
Department of Aquatic Animal Health Management
Faculty of Fisheries and Marine Science
Sher-e-Bangla Agricultural University, Dhaka, Bangladesh
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GitHub | ResearchGate | Google Scholar | ORCID | LinkedIn | University Portal



EDUCATION

Master of Science in Fish Health Management
Sylhet Agricultural University, Sylhet-3100, Bangladesh — July 2012-Dec 2013
Relevant Coursework: Fish Immunology, Fish Parasitology, Epidemiology, Research Methodology

Bachelor of Science in Fisheries
Sylhet Agricultural University, Sylhet-3100, Bangladesh — Jan 2008 – Dec 2011
Relevant Coursework: Fish Pathology, Fish Pharmacology, Fish Immunology, Microbiology, Statistics, Chemistry, Biochemistry, Fish Breeding and Genetics, Fundamentals of Molecular Biology and Biotechnology

TEACHING EXPERIENCE

Assistant Professor — 12/2018 - Current
Department of Aquatic Animal Health Management, Faculty of Fisheries & Marine Science
Sher-e-Bangla Agricultural University, Sher-e-Bangla Nagar, Dhaka-1207, Bangladesh
Master's Course Conduct: Bioinformatics and Computational Biology, Viral Diseases of Fish & Shellfish, Fish Immunology and Vaccine Development, Molecular Fish Parasitology
Undergraduate Course Conduct: Fish Pathology and Immunology

Assistant Professor — 10/2015 - 12/2018
Department of Fish Health Management, Faculty of Fisheries
Sylhet Agricultural University, Sylhet-3100, Bangladesh
MS Course Conducted: Advanced Fish Parasitology, Advanced Fish Immunology, Aquatic Pathobiology
Undergraduate Course Conducted: Fish Pathology, Fish Parasitology, Fish Immunology

Lecturer — 10/2013 - 10/2015
Department of Fish Health Management, Faculty of Fisheries
Sylhet Agricultural University, Sylhet-3100, Bangladesh
Undergraduate Course Conducted: Fish Pathology, Fish Parasitology, Fish Immunology

TECHNICAL SKILLS

Molecular lab work

- Experimental design, setup, management of laboratory workflows and biological studies
- DNA extraction, PCR amplification, gel electrophoresis documentation and sequence library preparation
- RNA extraction, cDNA synthesis, RT-PCR, and qPCR quantification and sequence library preparation
- Bacterial culture, biochemical characterization, antibiogram testing, and molecular identification
- Fish parasite morphometric identification, permanent slide preparation, and molecular identification

Bioinformatics skills

- **Programming and Computational Environment:** Python, R, and Bash for biological data analysis and pipeline development; workflow management using Nextflow; version control with Git and GitHub; package and environment management using Conda, Pixi and Docker; development using Google Colab, Kaggle and Visual Studio Code; high-performance computing in UNIX/Linux systems and cloud platforms including Amazon Web Services and Google Cloud.
- **Genomics and Variant Analysis:** sequence quality assessment and preprocessing (FastQC, MultiQC, Trimmomatic, Cutadapt); genome alignment and mapping using BWA and Bowtie2; variant discovery and structural variant analysis using GATK, FreeBayes, BCFtools, Manta, and CNVkit; genome annotation and functional variant interpretation with Annovar, SnpEff, and VEP.

- **Transcriptomics (Bulk and Single-Cell RNA-seq):** read alignment and transcript quantification using STAR, HISAT2, Salmon, Kallisto, and featureCounts; differential gene expression analysis with DESeq2 and edgeR; single-cell matrix data clustering, integration, and trajectory analysis using Seurat, Scanpy, Harmony, SingleR, and Monocle. miRNA profiling and analysis.
- **Bacterial Whole-Genome Analysis:** genome assembly and quality assessment using SPAdes, Unicycler, and QUAST; genome annotation with Prokka and RAST; antimicrobial resistance and gene detection using ABRicate, ResFinder, and CARD; sequence typing using MLST.
- **Metagenomics and Microbiome Analysis:** amplicon and shotgun metagenomic profiling using QIIME2 and DADA2; taxonomic classification with Kraken2, Bracken, and MetaPhlAn; functional pathway analysis using HUMAnN; downstream ecological and diversity analysis using MEGAN, Phyloseq, and PICRUSt2.
- **Environmental DNA and Metabarcoding:** biodiversity assessment using QIIME2, DADA2, and Vsearch for sequence clustering, taxonomic assignment, biodiversity and community profiling.
- **Mitochondrial Genome Analysis:** Complete mitochondrial genome assembly and characterization of Lates calcarifer using GetOrganelle, SPAdes, and MITOS2, supported by FastQC quality evaluation and downstream analyses including nucleotide composition, codon usage, phylogenetics, and control region characterization.
- **Epigenomic Analysis:** genome-wide DNA methylation analysis including quality control, alignment of bisulfite sequencing reads, methylation calling using Bismark, and identification of differentially methylated regions (DMRs) using methylKit and DSS. Chip-seq data analysis.
- **Proteomics and Structural Bioinformatics:** protein structure modeling and visualization using SWISS-MODEL, PyMOL, and Discovery Studio; molecular docking and virtual screening with PyRx and AutoDock Vina; structural pocket analysis using CASTp; ligand and drug-likeness and pharmacokinetic prediction using SwissADME; structural data retrieval from RCSB PDB; molecular dynamics simulation and structural stability analysis using GROMACS.
- **Population and Quantitative Genomics:** estimation of population differentiation (FST) and haplotype diversity; genome-wide association studies using PLINK, TASSEL, and GAPIT; QTL mapping and linkage analysis using MapQTL, QTL Cartographer, and R packages for genetic mapping.
- **Geospatial Analysis:** geographic data visualization and spatial mapping using QGIS and ArcGIS.

BIOINFORMATICS WORKFLOW

- **Bulk RNA-Seq Meta-Analysis Pipeline (Omyk_Stress_MetaRNASeq).** RNA-seq preprocessing and meta-analysis scripts for heat- and hypoxia-stress responses in rainbow trout (Oncorhynchus mykiss). [Public]. GitHub: https://github.com/BariSayed/Omyk_Stress_MetaRNASeq
- **Shotgun Metagenome Analysis Pipeline.** Pipeline for shotgun metagenome analysis. [Public]. GitHub: https://github.com/BariSayed/Shotgun-metagenomic-analysis-pipeline
- **Single-cell RNA-seq Meta-Analysis (Zebrafish Heart Injury).** Meta-analysis of single-cell transcriptomic responses to heart injury in zebrafish; data, scripts, figures and manuscript. [Private]. GitHub: https://github.com/BariSayed/zebrafish-heart-scrnaseq-meta
- **DNA Methylation Meta-Analysis (Rice Stress Methylome).** Harmonized WGBS meta-analysis of rice (Oryza sativa) DNA-methylation responses to biotic and abiotic stress. [Private]. GitHub: https://github.com/BariSayed/rice-stress-methylome-meta
- **Genome-wide Meta-QTL Analysis (Rainbow Trout).** Meta-QTL analysis of consensus genomic regions for disease resistance, growth and production traits in rainbow trout. [Private]. GitHub: https://github.com/BariSayed/mqtl_rainbow_trout
- **Aeromonas veronii Whole-Genome Characterization (SAUAV2).** Whole-genome characterization of Aeromonas veronii isolated from diseased tilapia in commercial aquaculture, Bangladesh. [Private]. GitHub: https://github.com/BariSayed/Aeromonas-veronii-SAUAV2
- **Streptococcus agalactiae One-Health Pangenome.** Comparative pangenome of human and aquaculture S. agalactiae; integrative/conjugative elements (ICEs) as major carriers of AMR asymmetry. [Private]. GitHub: https://github.com/BariSayed/streptococcus-agalactiae-onehealth-pangenome
- Additional workflows: 16S rRNA-based gut-microbiome analysis; bacterial whole-genome analysis; complete mitochondrial genome assembly, annotation and phylogenetics.

RESEARCH EXPERIENCE

Principal Investigator 07/25-06/26

The complete mitochondrial genome sequence of white leg shrimp, *Penaeus vannamei* of Bangladesh, Sher-e-Bangla Agricultural University Research System (SAURES), Sher-e-Bangla Agricultural University, Bangladesh

Principal Investigator 07/25-06/26

Metagenomic Analysis of Gut Microbiome in Invasive Sucker Fish (*Hypostomus plecostomus*) from the Polluted Buriganga River, Bangladesh, Research Grant, University Grants Commission of Bangladesh

Principal Investigator 07/25-06/26

The complete mitochondrial genome sequence of Asian Seabass, *Lates calcarifer* of Bangladesh, Fund from Research & Development Project, Ministry of Science and Technology (MOST), Bangladesh

Principal Investigator 07/24 -06/25

Molecular characterization and antibiotic resistance profiling of shrimp acute hepatopancreatic necrosis disease (AHPND) causing *Vibrio parahaemolyticus* from the southwestern shrimp farms of Bangladesh. Fund from Research & Development Project, Ministry of Science and Technology (MOST), Bangladesh

Principal Investigator 07/24-06/25

Morphometric and molecular identification of helminth parasites isolated from freshwater snakeheads (*Channa* spp.) in Bangladesh, Sher-e-Bangla Agricultural University Research System (SAURES), Sher-e-Bangla Agricultural University, Bangladesh

Principal Investigator 07/23-06/24

Molecular identification and antibiotic susceptibility profiling of pathogenic bacteria isolated from *Oreochromis niloticus* in Bangladesh. Fund from Research & Development Project, Ministry of Science and Technology (MOST), Bangladesh

Principal Investigator 07/21-06/22

Isolation, identification, and prevalence of parasites from Dhaleshwari River, Bangladesh. Fund from Research & Development Project, Ministry of Science and Technology (MOST), Bangladesh

Principal Investigator 07/20-06/21

GIS modelling of potential aquacultural sites in Dhaka Division. Fund from Research & Development Project, Ministry of Science and Technology (MOST), Bangladesh

Principal Investigator 07/17-06/18

Identification of pathogenic bacteria from diseased stringing catfish *Heteropneustis fossilis* with their sensitivity to antibiotics. Fund from Sylhet Agricultural University Research System, Sylhet Agricultural University, Sylhet, Bangladesh

Principal Investigator 07/15-06/16

Present status of fish diseases and their impact on the different aquacultural farms in Sylhet division. Fund from Sylhet Agricultural University Research System, Sylhet Agricultural University, Sylhet, Bangladesh

Principal Investigator 07/14-06/15

Effect of parasitism on the growth of *Heteropneustes fosillis.* Fund from Research & Development Project, Ministry of Science & Technology (MOST), Bangladesh

PUBLICATIONS (denotes corresponding author) *

1. Zewdie MC, **Bari SM***, Ahmed ST, Hamid MA, Bhuyain MAB, Ali SM, Magondu EW, Rossignoli CM, Yossa R. 2026. Market dynamics, consumer preferences, systemic barriers, and the potential of integrated multi-trophic aquaculture in Bangladesh. Frontiers in Sustainable Food Systems. 10: 1781267. https://doi.org/10.3389/fsufs.2026.1781267 **(Q1, IF: 4.1)**
2. **Bari SM*,** Fuad M, Hossain MJ, Ahammad I, Begum M, Helal MMU. 2025. A meta-analysis of public RNA-Seq data identifies conserved stress responses in Rainbow trout. BMC Genomics. 26: 999. https://doi.org/10.1186/s12864-025-12127-2 **(Q1, IF: 3.9)**
3. Begum M, Barsha KF, Rahman MM, Sarkar MMH, Chowdhury SF, Shormi AS, and **Bari SM***. 2025. Gut microbiome profiling of antibiotic-treated *Mystus cavasius* using culture- based and shotgun metagenomic approaches. Antonie Van Leeuwenhoek. 118(2): 183. https://doi.org/10.1007/s10482-025-02195-5 **(Q3, IF: 2.3)**
4. Hossain MJ, Rasin JH, Altaib H, Shyama IJ, Shawon RAR, Badr Y, Sohidullah M, Sakyi ME, **Bari SM,** Iqbal MM, Rahman MM. 2025. Prevalence and antibiotic resistance patterns of *Vibrio cholerae* isolated from diverse food sources in Khulna, Bangladesh. Food and Humanity. 100653. https://doi.org/10.1016/j.foohum.2025.100653 **(Q3, IF: 3.5)**
5. **Bari SM*,** Ahammad I, Bhattacharjee A, and Sahabuddin AM. 2025. Whole genome sequence data of *Aeromonas diversa* SAU1 isolated from Tilapia (*Oreochromis niloticus*). Data in Brief. 111396. https://doi.org/10.1016/j.dib.2025.111396 **(Q3, IF: 1.4)**
6. **Bari SM***, Marma M, Reja NB, Hossain A, Ahmed SKF, Rayhan, N, Islam MS, Hossain MA, Raman MM, and Alom MS. 2024. In-silico identification of Indian gooseberry (*Emblica officinalis*) compounds inhibiting

thermolabile hemolysin from *Vibrio alginolyticus* in Shrimp. Current Computer-Aided Drug Design. 21(2). https://doi.org/10.2174/0115734099342492250120114644 **(Q3, IF: 1.6)**

7. Jahan F, Rahman MH, Ahammad B, Khandaker M, Rahman MZ, Rasel M, Nafis R, Shaila A, **Bari SM**, Chandra MB. 2025. *Argulus japonicus*: A concerning health issue for *Channa punctata* collected from south-western Bangladesh. Veterinary Medicine and Science. 11(2). 11:e70250. https://doi.org/10.1002/vms3.70250 **(Q1, IF: 1.7)**
8. **Bari SM***, Reza NB, Marma M, Ahmed SKF, Hossain MA, Jabed MN, Hossain M.A., Monzoor M., and Alom M.S. 2025. Computational analysis of *Allium sativum* compounds to identify thermolabile hemolysin inhibitors against *Vibrio alginolyticus* in Shrimp. Journal of Advanced Biotechnology and Experimental Therapeutics. 8(1): 01-20. https://doi.org/10.5455/jabet.2025.01 **(Q4, CS: 2.2)**
9. **Bari SM*,** Islam MM, Amina A, Khatun MM, and Sahabuddin AM. 2024. Molecular identification, histopathology, and antibiotic susceptibility profiling of *Aeromonas veronii* isolated from *Oreochromis niloticus* in Bangladesh. Veterinary Medicine and Science, 10(6): e70103. https://doi.org/10.1002/vms3.70103 **(Q1, IF: 1.7)**
10. **Bari SM*,** Rahman MM, Amina A, Nadia ZM, Shawon RAR, Rahman M.M., Masum M.M. and Habib K.A. 2024. Morphological and molecular identification of *Euclinostomum heterostomum* (Trematoda: Clinostomidae) from spotted snakehead *Channa punctata* in Bangladesh. Systematic Parasitology. 101:62. https://doi.org/10.1007/s11230-024-10175-y **(Q3, IF: 1.2)**
11. **Bari SM*,** Hossen MI, Islam MR, and Helal MMU. 2024. Genome-wide identification and characterization of Interleukin-18 gene family in Rainbow trout (*Oncorhynchus mykiss*). Journal of Advanced Biotechnology and Experimental Therapeutics. 7(2): 275-289. https://doi.org/10.5455/jabet.2024.d23 **(Q4, CS: 2.2)**
12. Hossen, MK, Bari SM, Barman PP, Roy R, and Das PK. 2022. Application of Python OpenCV to detect contour of shapes and colour of a real image. International Journal of Novel Research in Computer Science and Software Engineering, 9, 20-25. https://doi.org/10.5281/zenodo.6576264
13. Deb A, Chandan S, Roy P, Hossain MI, **Bari SM**. 2021. Inland aquaculture and fish health management: a case study of Sylhet district in Bangladesh. Aquaculture Studies. 21: 129-137. http://doi.org/10.4194/2618-6381-v21_3_05 **(Q4, CS: 2.2)**
14. Goni O, Alam MMM, Khalil SMI, **Bari SM**, Hamom A, Parven M. and Mamun MAA. 2019. Identification of pathogenic bacteria from diseased stringing catfish *Heteropneustis fossilis* with their sensitivity to antibiotics. International Journal of Fisheries and Aquatic Studies. 8(1): 291-301. ISSN: 2347-512
15. Das H, Alam MMM, **Bari SM**, Mamun MA, Das M, Chowdhury NK, and Islam MS. 2018. Parasitic infestation in *Channa punctatus* at Sylhet in Bangladesh. Journal of Fisheries and Life Sciences. 3(2): 1-8. ISSN: 2456-6268
16. Mamun MAA, Nasren S and **Bari SM**. 2018. Role of probiotics in aquaculture: importance and future guidelines. J. Bangladesh Acad. Science. 42(1). 105-109. DOI: https://doi.org/10.3329/jbas.v42i1.37837
17. Ahammad B, Khandaker M, Hossain MI, Mamun MAA, Khalil SMI and **Bari SM**. 2017. Assessment of fish diversity in Shatghari point of Surma River, Golapgonj, Sylhet, Bangladesh. International Journal of Fisheries and Aquatic Studies. 5(5): 205-211. ISSN: 2347-5129
18. Mamun MAA, **Bari SM**, Das H, Islam S, Iqbal MM, and Nasren S. 2016. Parasitic infestation in *Channa punctatus* (Bloch 1793) collected from Lala Bazar fish market, Sylhet, Bangladesh. International Journal of Natural Science. 6(1): 49-53. ISSN: 2221-1020
19. **Bari SM***, Mamun MAA, Khalil SMI, Islam MJ, Baten MA and Hossain MM. 2015. Seasonal variation in population dynamics of helminth parasites in *Clarias batrachus* from natural wetland of Sylhet, Bangladesh. International Journal of Natural Science. 5(2): 86-89. ISSN: 2221-1020
20. Rahman MM, Alam MMM, Khalil SMI, **Bari SM** and Rashid MM. 2015. Status of chemicals and aqua drugs used in freshwater aquaculture in north-eastern Bangladesh. Journal of Sylhet Agricultural University. 2(2): 243-252. ISSN: 2308-1597
21. Hossain MM, Ahmed GU, Ferdous MJ, Khalil SMI, **Bari SM**, Hassan MM and Gosh K. 2014. Seasonal variation of clinicopathological changes of a farmed exotic carp *Hypophthelmichthys molitrix* from north-eastern region Bangladesh. Annals of Veterinary and Animal Science. 1(3): 122-135. ISSN: 2312-9123
22. **Bari SM**, Mamun MAA, Khalil SMI, Hossain MM, Marine SS and Hossain MM. 2014. Seasonal variation and effects of helminth infestation on *Clarias batrachus* (Linnaeus 1957) from the haor basin of Sylhet region of Bangladesh. Journal of Sylhet Agricultural University. 1(2): 231-237. ISSN: 2308-1597
23. Khalil SMI, Mamun MAA, **Bari SM** and Haque MN. 2014. Endoparasitic infestation in *Channa punctatus* (Bloch 1973) collected from different water bodies of Sylhet. Journal of Sylhet Agricultural University. 1(1): 59-64. ISSN: 2308-1597

IN REVIEW

1. **Bari SM*,** Khanum H, Reza NB, Marma M, Amina A. Habib KA. 2026. A checklist of helminth parasites in freshwater fishes of Bangladesh. Veterinary Medicine and Sciences **(Accepted) (Q1, IF: 1.7)**
2. **Bari SM***Marma M, Khan SN, Paul DC**,** Habib KA, Karim MM. 2026. First complete mitochondrial genome of Asian seabass (*Lates calcarifer*) from Bangladesh reveals Indian Ocean lineage affiliation and evolutionary relationships within Latidae. Aquaculture Report **(In Review) (Q1, IF: 3.8)** http://dx.doi.org/10.2139/ssrn.7133980

3. **Bari SM***, Imon MRI, Javeed MN, Mitu S, Islam T, Hossain MA, Islam MS 2026. Immunoinformatics design of a novel chimeric multiepitope vaccine against bacterial diseases in Pangas catfish (*Pangasius pangasius*). PLOS ONE. (**In Review**) **(Q1, IF: 3.2)**
4. Sarker BK, Begum M, **Bari SM**, Sultana N, Islam S, Chowdhury F, Kabir MA, Rabbane MG. 2026. Gut microbiota composition and nutritional quality of flathead grey mullet (*Mugil cephalus*) from riverine and gher aquaculture systems in coastal Bangladesh. Environmental Microbiology (**In Review**) (**Q2, IF: 4.0**)
5. Rahman MDMR, Shawon RAR, Haque MM, Sad AA, Sagor IH, Zaman R, Arnob FA, Naima MJ, Rasin JH, Rahman A, Paul TK, **Bari SM**, Hasan MT, Nadel AMMH, Abdullah AFM, Iqbal MM, Moribe J. 2026. Preparedness of students for zoonotic disease prevention in Bangladesh: A cross-sectional study of knowledge, attitudes and perceptions. Animals and Zoonoses. (**In Review**). **(Q1, IF: 4.2)**
6. **Bari SM***, Rahman MA, Mostafa F, Hossain MM, Rahman Z, Begum M, Helal MMU. 2026. A meta-analysis of single-cell transcriptomic (scRNA-seq) responses to heart injury in Zebrafish. BMC Genomics. (***In Review***)
7. **Bari SM***, Helal MMU, Zahangir MM. 2026. Genome-Wide Meta-QTL Analysis Identifies Consensus Genomic Regions for Disease Resistance, Growth, and Production Traits in Rainbow Trout (*Oncorhynchus mykiss*)
8. Helal MMU*, Israfil M, **Bari SM*** Shared and stress-specific DNA methylation programs across biotic and abiotic stresses in rice (*Oryza sativa*): a cross-subspecies multi-study meta-analysis. BMC Plant Biology (***In Review***).

CONFERENCE PRESENTATION AND AWARDS

- **Co-chair, Fish Disease Session,** Fisheries Society of Bangladesh Young Fisheries Students Congress (FSB YFSC-2025) held on 8th January 2026, Dhaka University, Dhaka, Bangladesh.
- **Best Oral Presenter**, Fish Disease session, 10th Biennial Conference of BFRF, December 2025, organized by Bangladesh Fisheries Research Forum (BFRF), Dhaka, Bangladesh
- **Best Oral Presenter**, Computational Biology and Bioinformatics session, International Conference of Biotechnology on Health and Agriculture (ICBHA), September 26-27, 2025, organized by Global Network of Bangladeshi Biotechnologists (GNOBB), Dhaka, Bangladesh
- **Best Oral Presenter**, Marine Biotechnology Session, International Conference on Biotechnology, June 20–21, 2025, organized by BRAC University, Dhaka, Bangladesh
- **Oral Presenter**, Fish Disease Session, Scientific Conference on Sustainable Aquaculture and Fisheries, April 29–30, 2025, organized by Chattogram Veterinary and Animal Sciences University, Chattogram, Bangladesh
- **Oral Presenter**, Aquaculture and Disease Session, International Fisheries and Aquaculture Conference, June 8, 2024, Department of Aquaculture, Rajshahi University, Rajshahi, Bangladesh
- **Oral Presenter**, International Conference on Advanced Agricultural Research, May 24–25, 2024, organized by Sylhet Agricultural University, Sylhet, Bangladesh

AWARDS, SCHOLERSHIP AND HONORS

- MS in Fish Health Management – Graduated 1st Class First (Sylhet Agricultural University, 2014)
- Bachelor of Fisheries Sciences – Top 10% of the class (Sylhet Agricultural University, 2012)
- Best Research Project Award – For outstanding study on fish disease management (Sher-e-Bangla Agricultural University, 2024)

PEER-REVIEW ACTIVITIES (Manuscript Reviewer)

- **Springer nature:** Aquaculture International, Scientific Reports, Journal of Parasitic Diseases
- **Wiley:** Reviews in Aquaculture, Journal of Fish Biology
- **Frontiers:** Frontiers in Endocrinology
- **Elsevier**: Heliyon

RESEARCH SUPERVISION

- Md. Muttakinul Islam (Master's supervision, 2025), Whole genome sequence of *Aeromonas* spp. collected from disease infected Parshe Fish (Grey Mullet).
- Meamaching Marma, (Master's supervision, 2025), Characterization of the complete mitochondrial genome of Asian seabass (*Lates calcarifer*) from Bangladesh.
- Nafees Bin Reza (Master's supervision, 2025), Gut microbial changes in healthy and disease affected Tilapia (*Oreochromis niloticus*)
- Rayhan-Ul Islam Imon (Undergraduate Thesis, 2024), Immunoinformatics design of a novel chimeric multiepitope vaccine against bacterial diseases in Pangas catfish (*Pangasius pangasius*).
- Md. Abdul Jabbar (Undergraduate Thesis, 2024), Morphometric and molecular identification of helminth parasites (*Euclinostomum heterostomum, Pallisentis* sp.) Collected from infected snakeheads (*Channa punctata*) in Bangladesh.

- Md. Sourove Islam (Master's supervision, 2016), Study on parasitic infestation in *Clarias batrachus* at Haor Region of Sylhet in Bangladesh.

TRAINING AND WORKSHOP ATTAINED (Online and Offline)

07/2026	2 days	QTL analysis, A two-day hands-on online training. Bruhaspathi Institute of Biosciences, India
07/2026	2 days	Chip-seq data analysis- A two-day hands-on online training. Bruhaspathi Institute of Biosciences, India
06/2026-	1 week	No Code and Agentic AI for Life Sciences. DeepBio academy, Bangladesh
06/2026	3 days	miRNA profiling- A three-day hands-on online training. Bruhaspathi Institute of Biosciences, India
05/2025	5 days	Nextflow Training Week 2025-Q1: Hands-on Bioinformatics Pipeline Development with Nextflow and nf-core, Seqera Community.
04/2026	5 days	Hand-on Training Chip-seq Data Analysis: from raw data to biological insights, Center for Biomedical Research and Training, India
04/2026	5 days	Hand-on Training on ATAC seq data analysis, Bruhaspathi Institute of Biosciences, India
03/2026	5 days	Hand-on Training on shotgun metagenomic data analysis, Bruhaspathi Institute of Biosciences, India
02/2026	2 days	Hand-on Training on eDNA metabarcoding data analysis, JSPS supported, Department of Fish Biology and Genetic, Sher-e-Bangla Agricultural University, Dhaka, Bangladesh.
02/2026	3 days	A five-day E-workshop on DNA methylation pipeline session, Bruhaspathi Institute of Biosciences, India
02/2026	5 days	Hand-on Training on metabolomic data analysis, Bruhaspathi Institute of Biosciences, India
01/2026	5 days	Hand-on Training on single-cell RNAseq data analysis using chronic kidney disease data, Bioinformatics and Genomics, University of Delhi, India
07/2025	5 days	Python and Biopython for Bioinformatics, Center for Biomedical Research and Training, India.
05/2025	12 days	Whole genome and exome sequencing analysis, Center for Biomedical Research and Training, India.
04/2025	12 days	Comprehensive bioinformatics for advanced microbial genomics, Oxford Discovery, London.
03/2025	5 days	Genome Wide Association Study data analysis, Center for Biomedical Research and Training, India.
01/2025	5 days	Microbial genomics: whole genome sequencing and metagenomic analysis, Decode life, India.
12/2024	5 days	Training on whole genome sequencing data analysis, Center for Biomedical Research and Training
09/2024	2 days	Python for Biologists, Institute of Biotechnological Research. India
08/2024	2 days	NGS fundamentals & data analysis: DNASeq-variant calling, OmicsLogics.com. India
07/2024	5 days	CRISPR-Cas9 gene editing mechanism, Microbiology society of Bangladesh, University of Dhaka.
05/2024	5 days	Microbial ecology: Amplicon and shotgun sequence-based metagenomics. DeCode Life, India
08/2023	12 days	Training on Bulk and Single cell RNA-seq data analysis with R, Chiral Bangladesh.
07/2023	2 days	Hand-on Training on Molecular taxonomy-DNA barcodes, Biosrishti, India.
06/2023	5 days	Hand-on Training on Hands-on practical RNA-seq data analysis with Linux and R, organized by BioCode.
05/2023	5 days	In silico vaccine design, Dawn of Bioinformatics, Dhaka, Bangladesh
04/2023	5 days	Computational drug design analysis, Dawn of Bioinformatics, Dhaka, Bangladesh
03/2023	5 days	Training on python for biologists, Chiral Bangladesh.
09/2022	5 days	RNA-seq data analysis using R programming, NextGenHelper, India
08/2022	5 days	Bioinformatics and genomics: Basics to advanced, NextGenHelper, India
03/2022	7 days	Linux and R, National Institute of Biotechnology, Dhaka, Bangladesh.
09/2019	7 days	R for data analytics, Institute of Statistical Research and Training, University of Dhaka, Bangladesh.

PROFESSIONAL MEMBERSHIP

- Member, Bangladesh Fisheries Research Forum (BFRF), Bangladesh
- Member, Zoological Society of Bangladesh (ZSB), Bangladesh
- Member, Bangladesh Society of Microbiologists (BSM), Bangladesh
- Member, American Society of Microbiology (ASM)
- Member, European Association of Fish Pathologists (EAFP)

- Member, Global Network of Bangladesh Biotechnologists (GNOBB), Bangladesh
- Member, International Society of Computational Biology (ISCB)
- Member, American Fisheries Society (AFS), United States
- Member, Krishibid Institute Bangladesh (KIB), Bangladesh
- Member, Sylhet Agricultural University Alumni Association (SAUAA), Bangladesh

LEADERSHIP EXPERIENCE

- Vice-president, Sylhet Agricultural University Blood Donation Club, Sylhet Agricultural University, Sylhet, Bangladesh, Jan 2010 - Dec 2010
- Vice-president, Fisheries Student Association, Sylhet Agricultural University, Sylhet, Bangladesh, Jan 2011 - Dec 2011
- Vice-president, Sylhet Agricultural University Science Club, Sylhet Agricultural University, Sylhet, Bangladesh, Jan 2012 - Dec 2012
- Assistant Secretary, Krishnochura Cultural Organization, Sylhet Agricultural University, Sylhet, Bangladesh, Jan 2013 - Dec 2013

ENGLISH PROFICIENCY (IELTS Score British Council Test)

Reading	Writing	Speaking	Listening	Overall
6.5	6.5	6.5	6.5	6.5

REFERENCES

1. Kazi Ahsan Habib, PhD
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